

Customer Churn & Transaction Analytics using SQL and Python

AI Business Report

CUSTOMER CHURN AND TRANSACTION ANALYTICS REPORT

1. EXECUTIVE SUMMARY

This report presents a comprehensive evaluation of customer behavior, churn dynamics, and revenue contributions for the organization. By analyzing historical transaction patterns and customer demographics using SQL and Python, the business identified critical operational bottlenecks and high-churn risk segments.

The analysis reveals an overall customer churn rate of 26.58 percent, representing 1,869 lost accounts out of a total base of 7,032 customers. While active customers generated over 16 million dollars in total revenue, short-term contract structures, sub-optimal onboarding experiences, and specific payment channels significantly degrade customer retention. Key operational variables such as tenure length, contract structure, and internet service type emerge as the primary determinants of churn.

Addressing these root causes through targeted contract incentives, enhanced first-year customer support, service quality upgrades, and ongoing metric tracking will mitigate revenue leakage and stabilize the long-term customer base.

2. PROJECT OBJECTIVE

The primary objective of this project is to analyze customer behavior patterns and transactional metrics to determine why customers terminate their services. By leveraging descriptive and inferential analytics, this project seeks to:

1. Identify the core drivers of customer churn across demographic, service, and account segments.
2. Evaluate revenue contribution patterns to pinpoint high-value customer cohorts.
3. Establish operational benchmarks and business recommendations aimed at lowering the overall churn rate while boosting long-term customer life value.

3. DATASET OVERVIEW AND METHODOLOGY

The dataset comprises 7,032 customer records evaluated across multi-dimensional parameters including account attributes, contract specifications, monthly charges, and total billing metrics.

Analytical Stack and Tooling:

- Database Management: SQL Server Management Studio (SSMS) for data extraction, joining, and aggregation.
- Data Processing and Analytics: Python environment utilizing Pandas for manipulation and cleaning.
- Data Visualization: Matplotlib within Google Colab for visual pattern identification.
- Analytical Support: Google Gemini AI for preliminary pattern validation and analytical optimization.

Workflow Architecture:

The analytics process progressed sequentially through data cleaning, SQL business queries, baseline Key Performance Indicator (KPI) extraction, customer behavior evaluation, revenue driver analysis, graphical trend plotting, and strategic recommendation drafting.

4. KEY PERFORMANCE INDICATORS

The quantitative baseline of the customer portfolio is outlined below:

Customer Volume Metrics:

- Total Customers: 7,032
- Active Customers: 5,163
- Churned Customers: 1,869
- Customer Retention Rate: 73.42%
- Customer Churn Rate: 26.58%

Financial Metrics:

- Total Revenue: \$16,056,168.70
- Average Revenue per Customer: \$2,283.73
- Average Monthly Charges: \$64.76
- Average Total Charges: \$2,283.73

Tenure Metrics:

- Average Customer Tenure: 32.42 Months

5. CUSTOMER BEHAVIOR ANALYSIS

The behavioral analysis highlights distinct variance across contract formats, customer lifetime milestones, service selections, and transaction channels.

Contract Structure Impact:

Month-to-Month contracts account for the overwhelming majority of churned accounts. In contrast, customers bound by One-Year and Two-Year commitments demonstrate substantially higher retention stability.

Tenure Breakdown:

The initial 12 months of service constitute the highest-risk period for customer attrition. Beyond the first year, customer churn drops progressively as tenure increases.

Service and Payment Type Influence:

1. Fiber Optic subscriber cohorts present disproportionately higher churn rates compared to DSL or Non-Internet subscribers, indicating potential service delivery or pricing friction points.
2. Electronic Check users exhibit an elevated rate of churn relative to automated credit card, bank transfer, or mailed check payment options.
3. Gender distribution across churned and retained segments shows negligible variance, proving that gender is not a meaningful predictor of churn.

6. REVENUE ANALYSIS

Financial evaluation demonstrates a clear alignment between operational longevity, contract structures, and maximum lifetime revenue generation.

1. Revenue Distribution by Contract Type: Two-Year contract holders contribute the highest total lifetime revenue and maintain the highest average billing totals over time.
2. Impact of Customer Tenure: Long-term accounts generate the largest share of overall company revenue, emphasizing that customer preservation directly correlates with long-term enterprise value.
3. Product Line Revenue: Fiber Optic services generate the highest absolute revenue across all product offerings despite higher churn rate dynamics, making it a critical revenue driver requiring maintenance.
4. Demographic Contributions: Non-Senior Citizen customers account for the majority share of total revenue, reflecting their higher overall representation within the active database.

7. CORRELATION ANALYSIS

Statistical relationships between continuous numerical variables yield critical operational insights:

1. Tenure versus Total Charges: Exhibits a strong positive correlation. Longer tenure predictably leads to higher total historical billings.
2. Monthly Charges versus Total Charges: Displays a moderate positive correlation. Monthly pricing influences total revenue, but customer longevity remains the dominant compounder.
3. Tenure versus Monthly Charges: Shows a weak correlation, demonstrating that pricing structures remain relatively uniform across short-term and long-term customer groups.

8. KEY FINDINGS

1. The critical churn window occurs within the first 12 months of customer onboarding.
2. Month-to-Month contracts are the primary operational vulnerability driving overall churn.
3. Fiber Optic subscribers yield the highest total revenue but experience unacceptable attrition rates due to potential service quality issues or cost sensitivity.
4. Electronic Check transaction methods correspond with higher customer loss rates compared to automated clearing house or card-based payment methods.
5. Customer retention directly correlates with financial performance, as Two-Year contract holders provide maximum financial stability.

9. BUSINESS RECOMMENDATIONS

1. Contract Conversion Incentive Programs:

Develop targeted marketing campaigns and modest pricing discounts to encourage customers on Month-to-Month arrangements to transition into One-Year or Two-Year agreements.

2. Onboarding Optimization in Year One:

Implement a specialized customer success program targeting the first 12 months of the customer journey. Provide proactive check-ins, tutorials, and dedicated support to reduce initial churn spikes.

3. Service Quality Enhancement for Fiber Optic Networks:

Conduct operational reviews of the Fiber Optic product line to diagnose network stability, competitive pricing models, and technical support response times to protect this high-revenue channel.

4. Automated Payment Migration Campaigns:

Encourage customers using Electronic Checks to adopt automated payment channels, such as recurring credit card or direct bank transfers, by offering small account credits.

5. Loyalty Reward Programs for Long-Term Accounts:

Establish structured loyalty programs to reward customers who surpass milestone tenures, reinforcing retention habits among high-value cohorts.

6. Metric Dashboard Implementation:

Build dynamic business intelligence dashboards to monitor monthly churn, contract conversions, and payment method shifts in real time for immediate intervention.

10. CONCLUSION

The quantitative analysis confirms that customer churn poses a actionable challenge to continuous revenue growth. While the overall business maintains a strong foundation of 5,163 active accounts generating over 16 million dollars in revenue, structural churn within Month-to-Month contracts and early tenure groups presents ongoing risk.

By executing the strategic recommendations focused on contract conversion, early-tenure onboarding support, Fiber Optic service improvement, and automated payment migration, the enterprise can systematically reduce its churn rate below the current 26.58 percent mark and secure predictable long-term profitability.